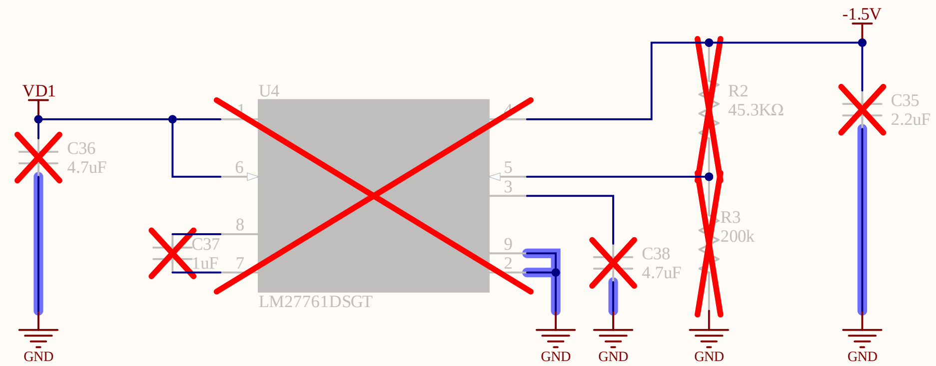
- Battery (VBAT)
- Always on 1.8V up to 750mA (1V8)
- System voltage between VBAT and VUSB (Vsys)
- Three software-controlled digital supplies (VD0, VD1, VD2)
- Two software-controlled analog supplies (VA0, VA1)

Default
- VD0: Off
- VD1: 1.8V or higher (Buck-Boost)
- VD2: Off
- VA0: Off
- VA1: 1.5 or 3.0V depending on config
- 1V8: Always On



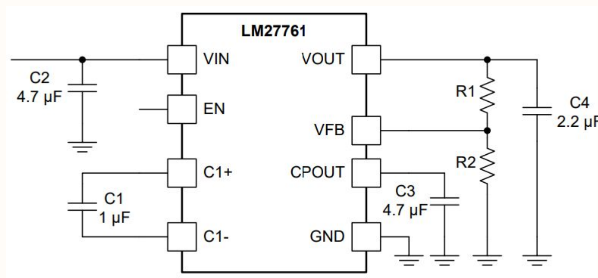


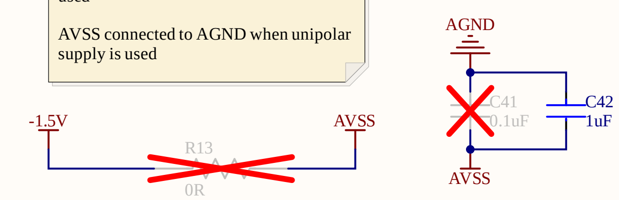
Figure 16. LM27761 Typical Application

$V_{OUT} = -1.22\ V \times (R1 + R2) / R2$

Analog supply for the ADS can be switched on/off with software controlled with the PMIC.

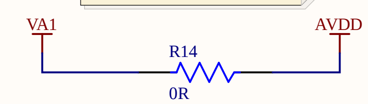
AVSS = -1.5V when bipolar supply is used

AVSS connected to AGND when unipolar supply is used

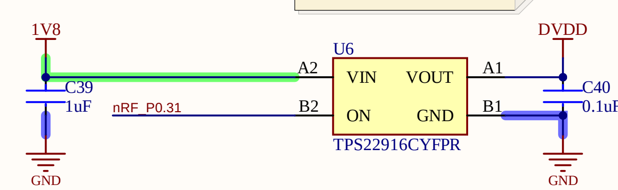


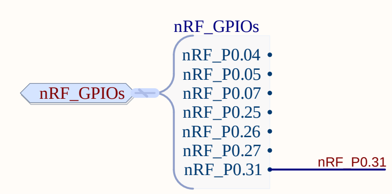
VA1 = 1.5V when bipolar supply is used

VA1 = 3.0V when unipolar supply is used

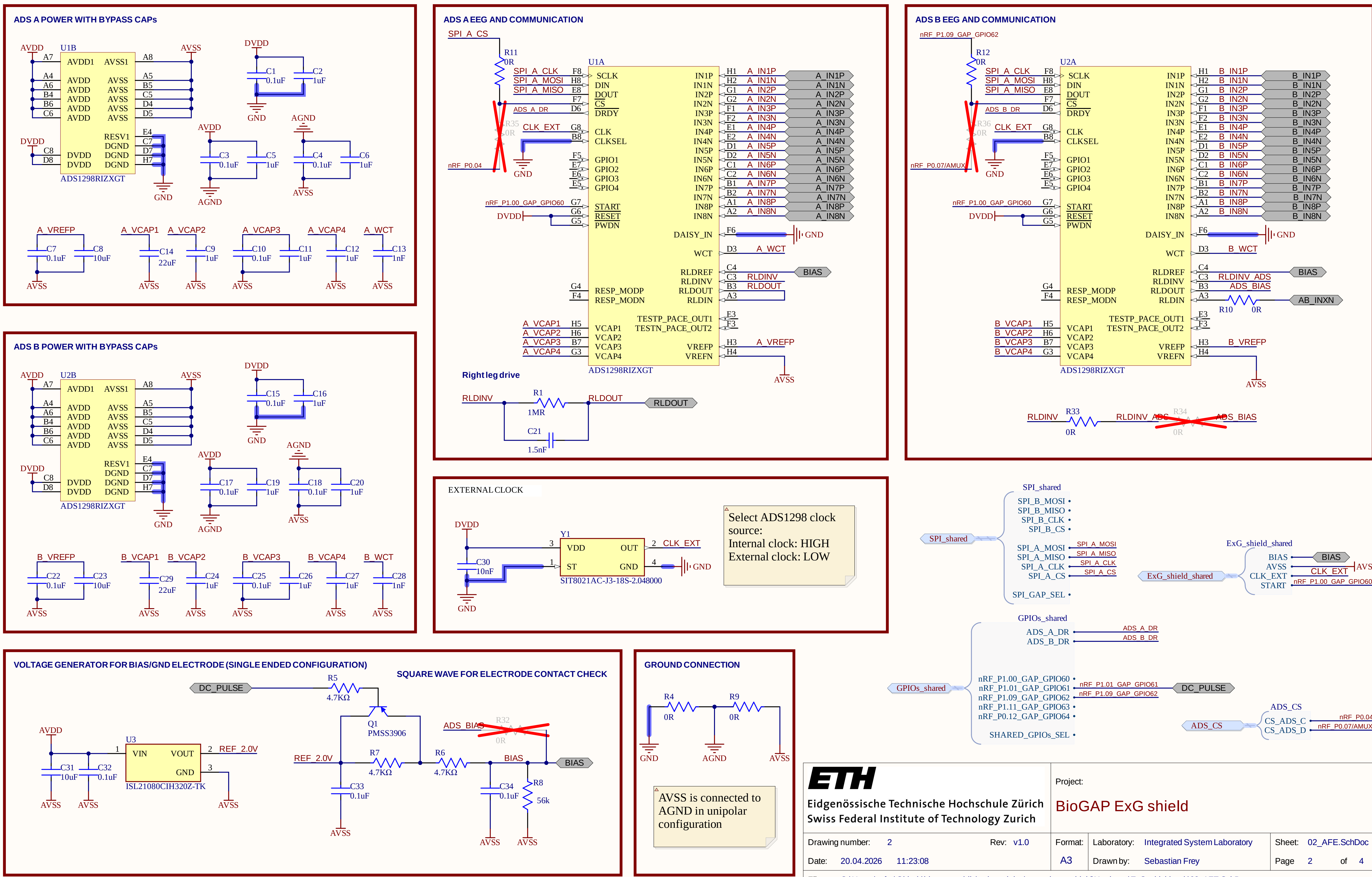


Digital supply for the ADS can be switched on/off with a load switch.





ETH Eidgenössische Technische Hochschule Zürich Swiss Federal Institute of Technology Zurich	Project: BioGAP ExG shield		
Drawing number: 1 Rev: v1.0	Format: A3	Laboratory: Integrated System Laboratory	Sheet: 01_power.SchDoc
Date: 20.04.2026 11:23:07		Drawn by: Sebastian Frey	Page 1 of 4
File: C:\Users\s-fre\Github\biogap_public\submodules\sensei-exg-shield\Hardware\ExG_shield_pcb\01_power.SchDoc			



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Project: BioGAP ExG shield

Drawing number: 2	Rev: v1.0	Format: A3	Laboratory: Integrated System Laboratory	Sheet: 02_AFE.SchDoc
Date: 20.04.2026 11:23:08			Drawn by: Sebastian Frey	Page 2 of 4

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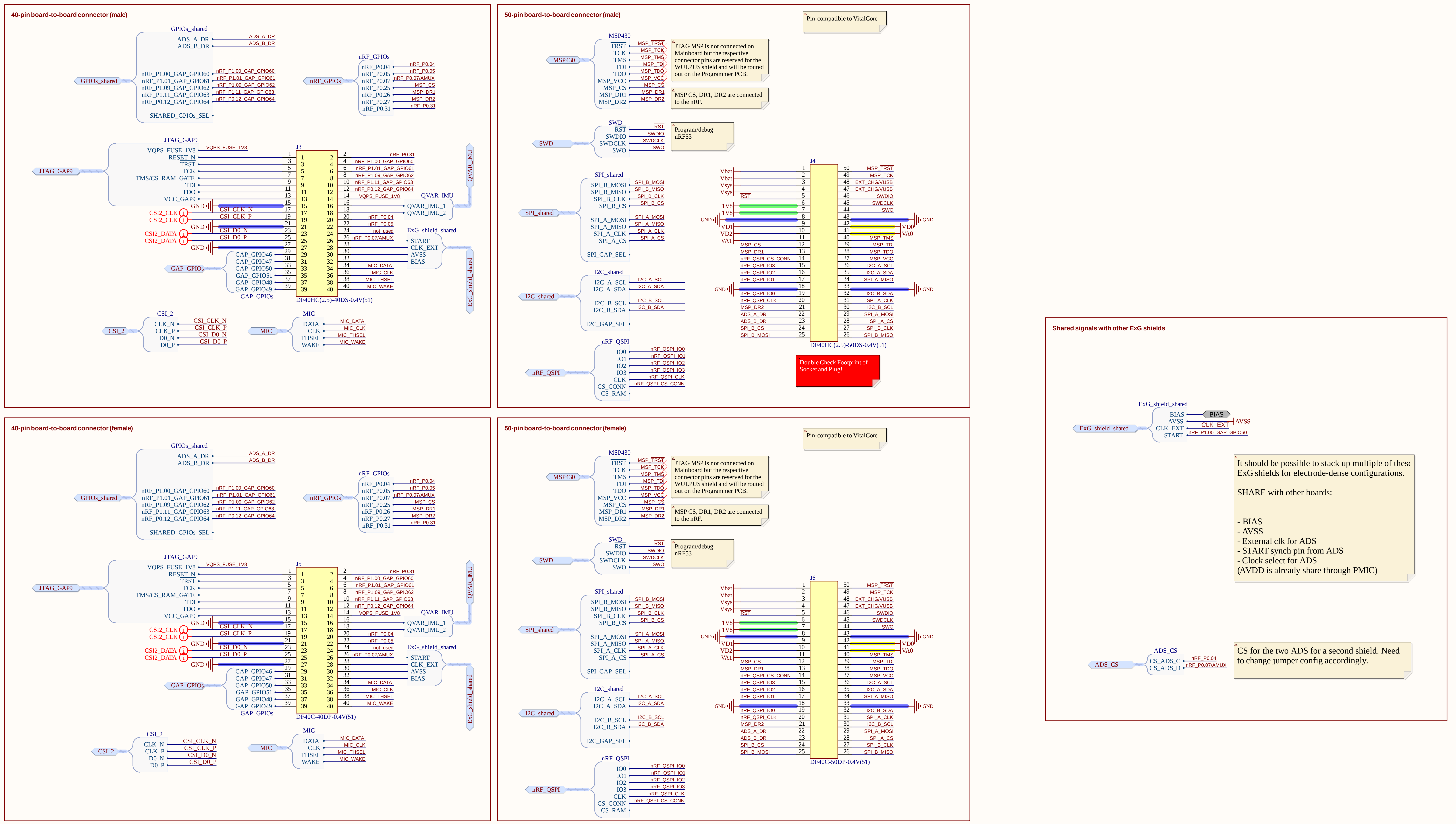

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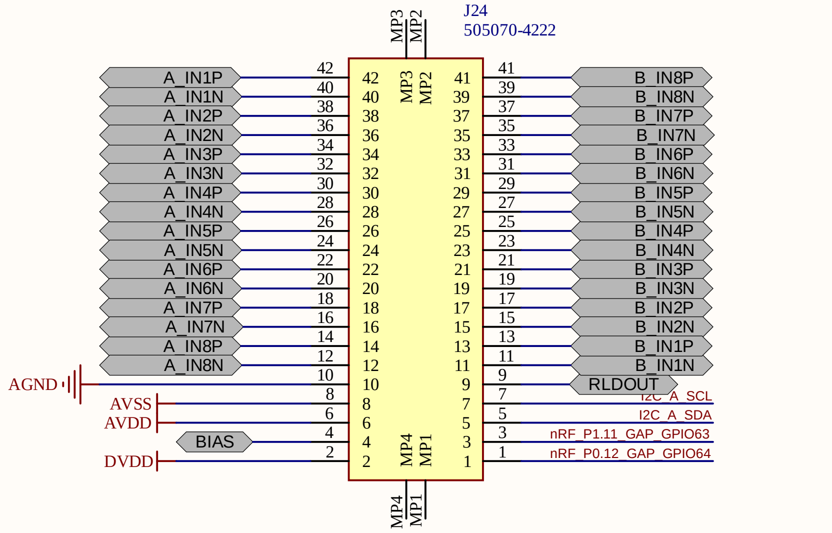
Drawing title: **BioGAP ExG shield**

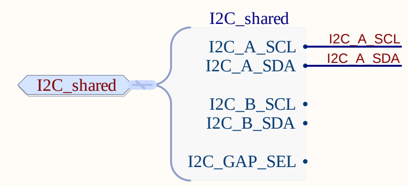
Drawing number: 4	Rev: v1.0	Format: A2	Laboratory: Integrated System Laboratory	Project: 04_gap_interface.SchDoc
Date: 20.04.2026 11:23:08		A2	Drawn by: Sebastian Frey	Page 4 of 4

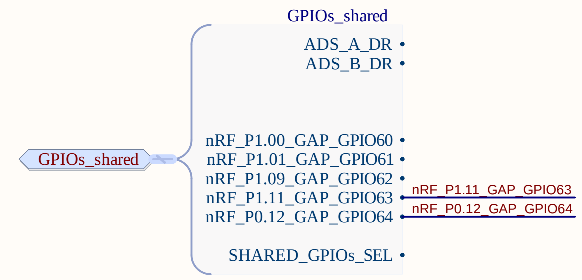
File: C:\Users\s-fre\Github\biogap_public\submodules\sensei-exg-shield\Hardware\ExG_shield_pcb\04_gap_interface.SchDoc

Board-to-board connector for fully differential config

For fully differential configuration we need:
- Fully differential channels (16 max)
- AGND which is used to set the body bias in case of bipolar supply
- BIAS in case of unipolar supply and pullups
- AVDD and AVSS (in case of active electrodes)
- I2C for future use
- Two shared GPIOs for future use





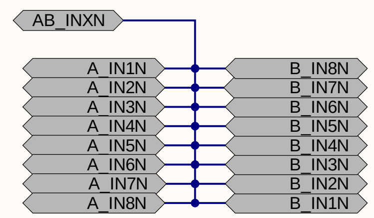


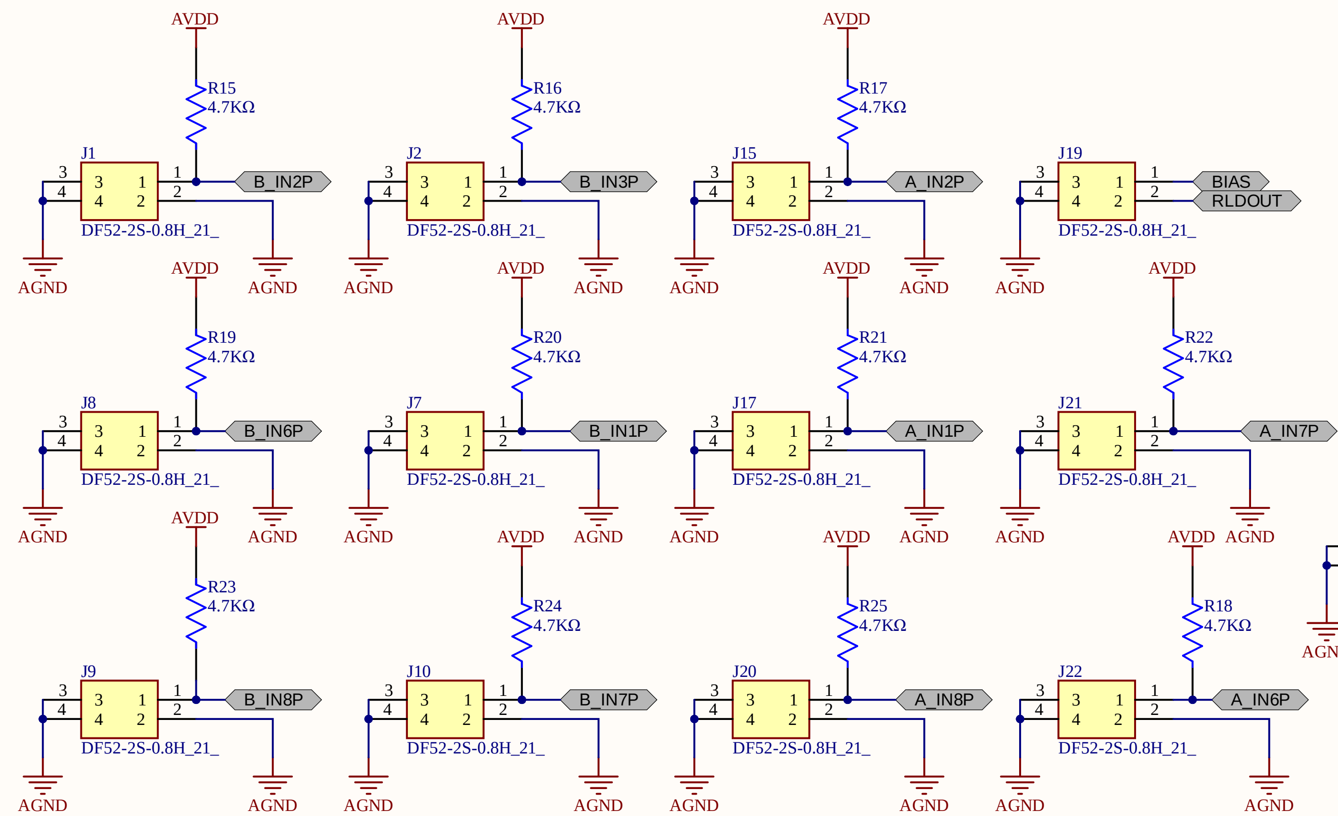
Break-off part

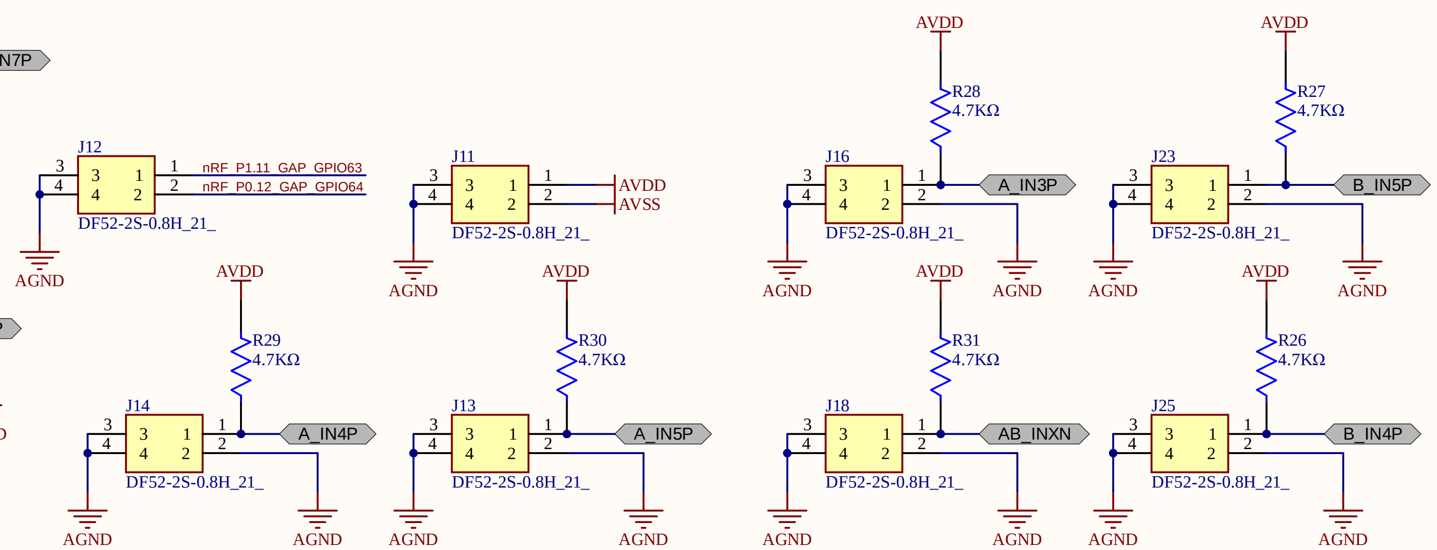
Connect together all neg ADS inputs for common reference. Only done on breakoff part of the PCB.

Breakoff part:
- Features a common REF for all negative inputs of the ADS and board-to-wire connectors.
- Can interface both passive and active electrodes (two-wire-config with pullups).
- Break this part away to use the fully differential configuration

For single ended configuration we need:
- Common reference (16 channels max)
- AGND which is used to set the body bias in case of bipolar supply
- BIAS in case of unipolar supply and pullups
- AVDD and AVSS (in case of active electrodes)
- Two shared GPIOs for future use







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Sheet: 03_electrode_interface.SchDoc		Page * of *	

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